

Assignment 7

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Complexity Analysis Python Program

(a) Time Complexity for Computing Top K Users

Experimental Observation

From the program outputs:

Number of Records (n)	Top K Calculation Time
1,000	~1.94 ms
10,000	~20.22 ms
100,000	~208.89 ms

Observation

When the number of activity records increases by **10×**, the execution time also increases by approximately **10×**.

This indicates that the algorithm grows **almost linearly with input size**.

Step-wise Operation Analysis

The computation of Top K users involves two major steps.

1. Computing total time per user

The function `total_time_per_user()` processes every log entry using `reduce()`.

For **n activity records**, each record is processed exactly once.

Therefore:

$$T_1(n) = O(n)$$

2. Sorting users based on total activity time

The totals dictionary contains **m unique users**.

Sorting is performed using:

```
sorted(totals.items(), key=lambda x: x[1], reverse=True)
```

Sorting complexity:

$$T_2(m) = O(m \log m)$$

Total Time Complexity

Combining both steps:

$$T(n) = O(n + m \log m)$$

Where:

- **n** = number of activity records
- **m** = number of unique users

Worst Case

If every activity record belongs to a different user:

$$m = n$$

Therefore:

$$T(n) = O(n + n \log n)$$

$$T(n) = O(n \log n)$$

Final Time Complexity

$$O(n + m \log m)$$

Worst case:

$$O(n \log n)$$

(b) Space Complexity of Storing Intermediate Results

The program stores several intermediate data structures.

1. Activity Logs

The list of activity records stores **n elements**.

$$S_1 = O(n)$$

2. Totals Dictionary

The dictionary stores total activity time for **m users**.

$$S_2 = O(m)$$

3. Sorted Users List

Sorting creates an additional list of **m elements**.

$$S_3 = O(m)$$

4. Unique Actions Set

The set stores **a unique actions**.

From the output:

Unique Actions = 6

Since this number is constant:

$$S_4 = O(1)$$

Total Space Complexity

$$S(n) = O(n + m + a)$$

Since **a is constant** and **m ≤ n**:

$$S(n) = O(n)$$

Final Space Complexity

$$O(n)$$

Final Conclusion

The analysis of the activity log analyzer and the experimental results obtained from program execution indicate the following:

- **Computing the total activity time per user** requires processing each activity record once. Therefore, its time complexity is $O(n)$, where n is the number of activity records.
- **Computing the Top K most active users** involves sorting the users based on their total activity time. If m represents the number of unique users, the sorting step has a time complexity of $O(m \log m)$. Hence, the overall complexity for determining the Top K users becomes $O(n + m \log m)$.
- The **additional space required** by the algorithm includes storing the totals dictionary and other intermediate structures. The totals dictionary requires $O(m)$ space, while the list of activity records requires $O(n)$ space. Since $m \leq n$, the overall space complexity is $O(n)$.

These observations are consistent with the experimental outputs, where the execution time increases proportionally with the number of activity records, confirming the theoretical complexity of the algorithms used in the program.

```
Enter number of activity records to generate: 1000

===== Activity Log Analysis =====

Total Activity Records : 1000
Unique Users           : 619
Unique Actions         : 6

Top 5 Most Active Users:
1. CSB24561
2. CSB24666
3. CSB24873
4. CSB24397
5. CSB24319

Performance Statistics
Total Time Calculation -> 1.5639 ms | 40.4375 KB
Top Users Calculation  -> 1.9449 ms | 75.2812 KB
Unique Actions         -> 0.0440 ms | 0.7578 KB

=====
```

```
Enter number of activity records to generate: 10000

===== Activity Log Analysis =====

Total Activity Records : 10000
Unique Users           : 6305
Unique Actions         : 6

Top 5 Most Active Users:
1. CSB29063
2. CSB24719
3. CSB31120
4. CSB29520
5. CSB32116

Performance Statistics
Total Time Calculation -> 13.1286 ms | 553.7031 KB
Top Users Calculation  -> 20.2270 ms | 842.7969 KB
Unique Actions         -> 0.5245 ms | 0.7578 KB

=====
```

Enter number of activity records to generate: 100000

===== Activity Log Analysis =====

Total Activity Records : 100000
Unique Users : 63254
Unique Actions : 6

Top 5 Most Active Users:

1. CSB119724
2. CSB79763
3. CSB104876
4. CSB38416
5. CSB119597

Performance Statistics

Total Time Calculation -> 102.2393 ms | 5237.7891 KB
Top Users Calculation -> 208.8915 ms | 8300.6172 KB
Unique Actions -> 3.0560 ms | 0.7578 KB

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